

Cell Mechanics Training App

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Context

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Basic Agent
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Cell Definition

Global Variables

User Parameters

PhysiCell Constants

List of cells

SVG options

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Default Cell Definition

Default microenvironment

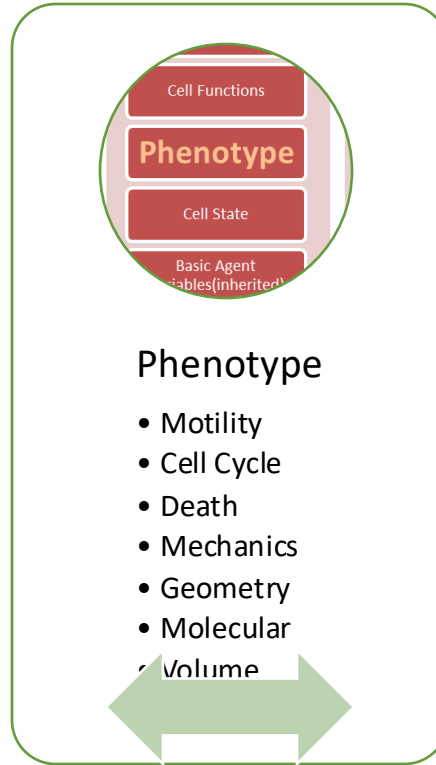


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Context



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Background

- Each Cell in Physicell can have its own parameters for the mechanics that dictate its motion.
- These parameters are given as a multiple of their original predicted strength.
- Therefore, setting a parameter to 0 “turns it off.”
- The mathematics are performed by updating the cell’s velocity *using* these parameters. The position is then updated with the second order Adam’s Bashforth discretization.

$$\mathbf{x}_i(t + \Delta t_{\text{mech}}) = \mathbf{x}_i(t) + \frac{1}{2} \Delta t_{\text{mech}} (3\mathbf{v}_i(t) - \mathbf{v}_i(t - \Delta t_{\text{mech}}))$$



Mathematics: Variables and Definitions

- **Cell to Cell Adhesion**

- A multiplier of cell to cell adhesion; it can be found on Slide 8 of this deck
“Mathematics: Models.” The next 3 listed parameters follow the same convention.

- **Cell to Cell Repulsion**

- **Cell to BM Adhesion**

- “BM” → Basement Membrane

- **Cell to BM Repulsion**

- **Relative Maximum Adhesion Distance**

- Maximum distance a multiplier of *geometry.radius*



Mathematics: Models

- The parameters here are simply multipliers for the existing models of adhesion and cohesion.

$$\mathbf{v}_i = \sum_{j \in \mathcal{N}(i)} \left(\overbrace{-\sqrt{c_{cca}^i c_{cca}^j} \nabla \phi_{1, R_{i,A} + R_{j,A}}(\mathbf{x}_i - \mathbf{x}_j)}^{\text{cell-cell adhesion}} - \overbrace{\sqrt{c_{ccr}^i c_{ccr}^j} \nabla \psi_{1, R_i + R_j}(\mathbf{x}_i - \mathbf{x}_j)}^{\text{cell-cell repulsion}} \right) - \overbrace{c_{cba}^i \nabla \phi_{1, R_{i,A}}(-d(\mathbf{x}_i) \mathbf{n}(\mathbf{x}_j))}^{\text{cell-BM adhesion}} - \overbrace{c_{cbr}^i \nabla \psi_{1, R_i}(-d(\mathbf{x}_i) \mathbf{n}(\mathbf{x}_j))}^{\text{cell-BM repulsion}} + \mathbf{v}_{i, \text{mot}}$$



Class structure: Data

```
class Mechanics
{
public:
    double cell_cell_adhesion_strength;
    double cell_BM_adhesion_strength;
    double cell_cell_repulsion_strength;
    double cell_BM_repulsion_strength;

    double relative_maximum_adhesion_distance;
    double maximum_adhesion_distance;

    void set_relative_maximum_adhesion_distance( double new_value );
    void set_relative_equilibrium_distance( double new_value );

    void set_absolute_equilibrium_distance( Phenotype& phenotype, double new_value );

    Mechanics();
};
```



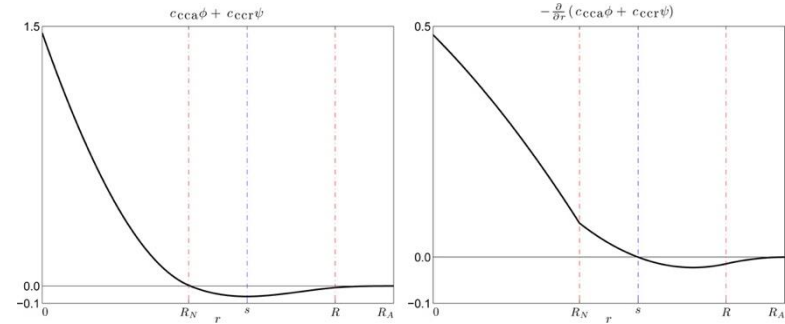
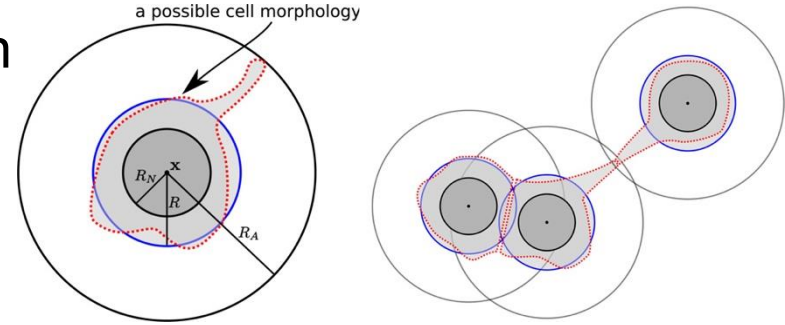
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Phenotype: Mechanics

- **Mechanics** keeps parameters for adhesion and "repulsion"
 - Key parameter: maximum adhesion distance
 - a multiple of the cell's radius
 - (as a multiple of the cell's radius)
- Default model uses potential functions, but this can be supplemented or replaced.



Documentation: User Guide 11.5



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Cell definition: mechanics

- **Options** give you some easy ways to *override* the cell-cell adhesion strength to accomplish other calibration goals:
 - **set_relative_equilibrium_distance** lets you choose the equilibrium cell-cell spacing, as a multiple of the cell radius. It will automatically choose a cell_cell_adhesion_strength to meet your selected equilibrium spacing.
 - 2.0 would have an equilibrium spacing of 2 cell radii (radius of cell 1 + radius of cell 2). Don't exceed this!
 - 1.8 or 1.9 is more typical.
 - **set_absolute_equilibrium_distance** allows you to choose this equilibrium distance in absolute (dimensional) units. This may or may not make sense as the cell changes size!



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Credits

Lesson Planning: Paul Macklin

Slides: Aneequa Sundus, Kali Konstantinopoulos*, Mary Chen*, Furkan Kurtoglu

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PhysiCell Development:

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- National Cancer Institute (U01CA232137)
- National Science Foundation (1720625)

Training materials:

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